

Multiple Choice

1. **Answer:** A

Options: A) mechanical waves; B) sound waves; C) all waves; D) None of these; E) radio waves

Note: Correct option: A - mechanical waves

2. **Answer:** A

Options: A) Sound; B) Ultraviolet waves; C) Infrared waves; D) Microwaves; E) X-rays

Note: Correct option: A - Sound

3. **Answer:** A

Options: A) oval; B) spiral; C) elliptical; D) All of these; E) rectangular

Note: Correct option: A - oval

4. **Answer:** A

Options: A) $[(3 f_2) / 2]$; B) $(f_1 + f_2) / 2$; C) $(2 f_2) / 3$; D) $2 f_2$; E) $(3 f_2) / 3$

Note: Correct option: A - $[(3 f_2) / 2]$

5. **Answer:** C

Options: A) Magnetic attraction; B) vacuum polarization; C) interactions with the ionic lattice; D) Gravitational pull; E) Electrostatic attraction

Note: Correct option: C - interactions with the ionic lattice

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.6 ampere'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'by contracting changing gravitational potential energy into thermal energy'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** .003 miles. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** $-1.32 \times 10^{13}\text{C}$. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key